

Total no. of pages 2

Roll no. _____

B.Tech. II Semester

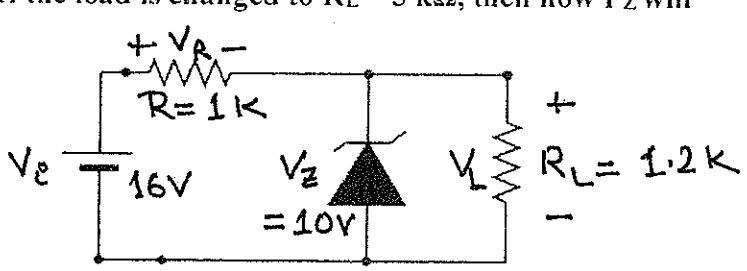
MID-SEMESTER EXAMINATION, February 2024

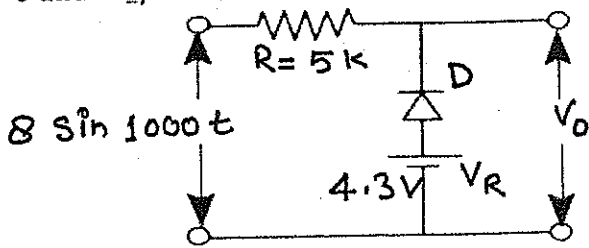
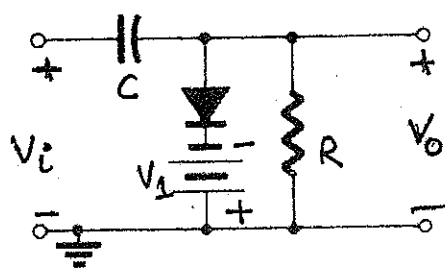
Course Codes : ECECC202 / EIECC202/ ICECC202 / ~~ECECC202~~
Course Title: Electronics Device and Circuits

Time: 1:30 Hours

Max. Marks: 20

Note: Attempt all questions. Missing data/information (if any) may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	
1a	Derive the expression for built in potential difference, $V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$ for an abrupt P-N junction.	2	CO1
1b	Find the current that flows in a silicon bar of 10- μ m length having a 5- μ m $\times$ 4- μ m cross-section and having free-electron and hole densities of 10^4 /cm ³ and 10^{16} /cm ³ , respectively, when a 1 V is applied end-to-end. Use $\mu_n = 1200$ cm ² /V.s and $\mu_p = 500$ cm ² /V.s	2	CO1
2a	State and derive Mass action law.	2	CO1
2b	A Zener diode regulator is shown in the Fig. 1 Determine V_L , V_R , I_Z and P_Z . If the load is changed to $R_L = 3$ k Ω , then how P_Z will change.  Fig.1	2	CO2
3a	Explain diffusion and transition capacitance in P-N diode. Does depletion capacitance increase or decrease with increase in magnitude of reverse bias voltage.	2	CO1

3b	Find the output voltage V_o and plot transfer curve of the clipper circuit shown in Fig.2, assuming that the diodes is modeled by $R_f = 0$ and $V_{\gamma} = 0.7$ V.  Fig.2	2	CO2
4a	Sketch V_o for clamper circuit shown in Fig.3 for the input signal $V_i = 10 \sin 1000t$, $V_1 = 5$ V assume that the diode is ideal and R is taken very large.  Fig.3	2	C02
4b	Sketch the circuit of Full wave Bridge rectifier and explain its working. Find V_{dc}, V_{rms} and efficiency of bridge rectifier.	2	CO2
5a	Determine the forward resistance (r_f) of a silicon PN junction diode when the forward current (I) is 7 mA at a temperature (T) of 300 K.	2	CO2
5b	Explain the working of <i>npn</i> transistor in active and saturation regions. Indicate the electron and hole current components for a <i>npn</i> transistor biased in the forward-active region with the help of diagram.	2	CO2